RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Department of Mechanical Engineering

Date: January 2025	**CIE – Improvement Test**	**Max. Marks: 50 + 10**
Semester: III	**UG**	**Duration: 120 Min**
Course Title: Material Science for Engineers		**Course Code: ME232TB**

S No	Questions	M	BT	CO
	Part A (Quiz)			
1	Mention the main purpose of annealing heat treatment process of ferrous materials.	1	1	3
2	In _____ type of annealing, the lamellar pearlite is transformed into globular type structure.	1	1	3
3	In pack (solid) carburizing _____ is used as energizer.	1	1	3
4	Give two examples for 1D nanomaterial.	2	1	3
5	What happens to the elastic modulus property, when the material brought to nanoscale?	1	1	4
6	Name two top-down methods of nano-material synthesis.	2	2	4
7	In sol-gel process the formation of an oxide network takes place through _____ reactions of a molecular precursor in the liquid.	1	1	4
8	Mesoporous materials are a class of nanomaterials characterized by their highly ordered porous structure with pore sizes ranging from _____ to _____ nanometers.	1	1	4
	Part B (Test)			
1a)	Describe the normalizing heat treatment process with the help of a neat sketch.	05	2	3
1b)	Discuss the flame hardening process with a neat sketch.	05	2	3
2	Illustrate the construction of TTT diagram for eutectoid steel of 0.8% carbon with neat sketches and label various features on the diagram. Also state the applications of the TTT diagram.	10	3	3
3a)	Enumerate the purposes of post-processing heat treatment methods of electronic devices.	05	3	3
3b)	Explain the rapid thermal processing with a neat sketch with its advantages.	05	3	3
4	"Identify whether the chemical vapour deposition (CVD) is top-down or bottom-up approach". Justify your answer and explain the working principle with the help of neat sketch. State its applications and advantages.	10	3	4
5	Discuss the working principle and various modes of atomic force microscope (AFM) with neat sketches.	10	3	4

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Department of Mechanical Engineering

Date: March 2024	**CIE – Improvement**	**Max. Marks: 50**
Semester: 3	**UG**	**Duration: 90 Min**
Course Title: Materials Science for Engineers		**Course Code: ME232AT**

S No	Questions	M	BT	CO
1a.	Define heat treatment and explain the purpose of heat treatment.	5	2	3
1b.	Discuss the advantages and applications of rapid thermal processing for electronic devices.	5	2	3
2a.	Differentiate between annealing and normalising heat treatment processes.	5	3	3
2b.	Describe the full annealing heat treatment process for ferrous materials.	5	3	3
3a.	Explain the steps involved in thermal oxidation of semiconductor devices.	6	3	3
3b.	Describe the defects of heat treatment process for different materials.	4	2	3
4.	With the help of TTT diagram explain hardening process.	10	3	3
5.	Explain the following heat treatment processes with neat sketches: i) Flame hardening and ii) carburising.	10	3	3

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Department of Civil Engineering

Date: 19-03-2024	**CIE – III (Test)**	**Max. Marks: 50**
Semester: III	**UG**	**Duration: 90 Min**
Course Title: Environment and Sustainability		**Course Code: CV232AT**

S No	Questions	M	BT	CO
1a	Enumerate the impacts of ozone layer depletion.	5	2	3
1b	Explain the concept of 5R.	5	2	3
2	Define the following. i) EIA ii) Climate change iii) Carbon Credit iv) Carbon Footprint v) Green Building.	10	2	3
3	Explain the principles of CSR.	10	2	4
4	Enumerate the ways to reduce carbon foot print.	10	3	3
5a	Briefly explain the concept of circular economy.	5	2	3
5b	Explain the concept of charity and corporate philanthropy.	5	2	4

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Department of Biotechnology

Date: 8th January 2025	**Improvement Test**	**Max. Marks: 60**
Semester: III	**UG**	**Duration: 120 Min**
Course Title: Bio Safety Standards and Ethics (Basket Course)		**Course Code: BT232AT**

S No	Questions (Quiz)	M	BT	CO
1	What is food processing?	1	1	3
2	What is Innovative Packaging? Give examples.	1	1	3
3	What are the advantages of preservation of food?	1	1	3
4	What are flavouring agents used in food? Give examples.	1	1	4
5	List out the health policies in India.	1	1	4
6	What is food safety?	1	1	4
7	What is ethics in food industry?	2	1	4
8	List out 4 methods used for food analysis.	2	1	4

S No	Questions (Test)	M	BT	CO
1	Classify food preservatives, explain the strategies used for various foods preservation.	10	2	3
2	Illustrate and classify on various food packaging methods used in food industry.	10	3	4
3	"Meat and meat products derived from livestock treated with antibiotics and hormones can cause adverse effects on consumer's health" justify the statement.	10	3	3
4	Elaborate on smart packaging methods with suitable examples. Add a note on packaging materials.	10	4	4
5	Write explanatory notes on: Clinical Ethics and Research Ethics.	5 + 5	2	3

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Department of Biotechnology

Date: 2023-24 (Odd Sem)	**Closed Book Offline Test – 3**	**Max. Marks: 50**
Semester: III	**UG**	**Duration: –**
Course Title: Bio Safety Standards and Ethics (Basket Course)		**Course Code: BT232AT**

S No	Questions (Test)	M	BT	CO
1	Explain on various food processing techniques.	10	2	4
2	Illustrate on the following: Blanching, Pasteurization, Frying.	10	4	4
3	Describe three levels of packaging the food products.	10	4	4
4	Explain the effects of poor packaging on food products, add a note on smart packing.	10	2	4
5	Write Explanatory notes on: Freezing, Dehydration, Homogenization.	10	2	4